

Douwe Rijs

Robotics & Machine Learning Engineer · MSc Robotics, TU Delft

Bosboom Toussaintplein 152, Delft, The Netherlands · +31 6 83903534 · Douwe@standofl.nl · linkedin.com/in/douwe-rijs-7990b2259

PROFILE

Robotics MSc student at TU Delft with a Mechanical Engineering background and a strong focus on machine learning, perception and control systems. Hands-on R&D experience in computer vision, transformer models and enterprise software development at Allseas. Motivated to apply advanced robotics and ML to real-world industrial systems.

EDUCATION

MSc Robotics — *Delft University of Technology* 2025 – present

- *Machine Learning*: SVMs, PCA, CNNs and RNNs applied to robotics problems.
- *Deep Reinforcement Learning*: value- and policy-based methods (DQN, actor-critic) for control and decision-making.
- *Machine Perception*: pose estimation from noisy sensor data; Kalman filtering; LiDAR, radar and stereo vision.
- *Robot Dynamics & Control*: impedance control and open-loop control of manipulators.
- *Planning & Decision Making*: sampling-based motion planning (RRT*), MPC, PID and search algorithms.

BSc Mechanical Engineering — *Delft University of Technology* 2022 – Oct 2025

- GPA 7.1/10; thesis graded 8.0/10.
- Thesis: chemotherapy-injection assisting device — medical-device regulations and CE certification of a clinical design.
- Core: mechanics, dynamics, materials science, control theory, numerical simulation.
- *Minor Computer Science*: machine learning fundamentals, camera geometry and 3D reconstruction, algorithms & data structures, software engineering and collaborative development (Git).

VWO (pre-university) — *Haarlemmermeerlyceum, Hoofddorp* 2016 – 2022

- Science & technology profile; languages: Dutch, English, Chinese.

EXPERIENCE

Assistant R&D Engineer — *Allseas, Delft* Feb 2025 – present

- Built a network observability platform polling 100+ switches via SNMP, aggregating port-level metrics (data rate, VLAN, link status) for real-time diagnostics — improving system insight and debugging efficiency.
- Developed a hybrid OCR/VLM pipeline that reads handwritten pipe IDs on curved steel surfaces; evaluated multiple transformer models and custom preprocessing, reaching ~81% accuracy on noisy industrial data.
- Implemented Azure DevOps CI/CD pipelines automating versioning and software releases from commits.
- Day-to-day development in Go, React/TypeScript and Python within an enterprise software process.

- Freelance service work at restaurants and events; client communication and own tax administration.

SELECTED PROJECTS

Greenhouse Monitoring Robot & Digital Twin (MIRTE) — *TU Delft, team of 5* 2026

- Autonomous ROS 2 robot on the MIRTE Master platform that maps a greenhouse with SLAM (Nav2, SLAM Toolbox), inspects every flower bed and streams detected plants, pests and climate data into a live digital twin.
- Owned the decision-making layer: a state machine orchestrating navigation, scanning, fault recovery and battery-triggered return-to-dock; YOLOv8 detection and AprilTag localisation validated in simulation and on the real robot.
- Managed the five-person team's GitLab issue board and branch/review workflow; 7 of 8 mandatory client requirements met in 8 weeks.

Autonomous Flight of a Micro Air Vehicle — *TU Delft MAVLab* 2025

- Vision-only obstacle avoidance for a Parrot Bebop 2 flying autonomously in the CyberZoo — no external motion capture or off-board compute.
- Implemented border, gate and vertical-obstacle detection in C/C++ inside the Paparazzi autopilot; optimised the vision stack to 8–11 FPS on the drone's onboard ARM processor.

Multi-Sensor 3D Object Detection — *TU Delft, Machine Perception* 2025

- Built a multi-sensor perception pipeline (LiDAR, camera, radar) for 3D bounding-box detection on the real-world View-of-Delft driving dataset, handling noisy and sparse sensor data.

Quadrotor Motion Planning — *TU Delft, Planning & Decision Making* 2025

- Implemented RRT* motion planning for autonomous quadrotor navigation and analysed its limitations in dynamic scenarios without integrated control (MPC).

TECHNICAL SKILLS

Programming	Python, Go, TypeScript/JavaScript, React, C++, MATLAB
Machine learning	CNNs, RNNs, Transformers, VLMs, deep reinforcement learning, SVM, PCA, Hugging Face, YOLO
Robotics	ROS 2, SLAM, Nav2, sensor fusion, Kalman filtering, LiDAR, radar, stereo vision, pose estimation
Control & planning	PID, MPC, impedance control, robot dynamics, RRT*, search algorithms
Tools & DevOps	Git, Azure DevOps, CI/CD, Docker, Linux, observability & monitoring

LANGUAGES

Dutch (native) · English (fluent) · Chinese (elementary)